Discussion: SMT and DELPH-IN

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What's going on in SMT?

Tiny incremental developments in traditional SMT got boring...

Let's use semantics!

- "A Discriminative Model for Semantics-to-String Translation"
  Aleš Tamchyna, Chris Quirk, Michel Galley
- "Language to Code: Learning Semantic Parsers for If-This-Then-That Recipes"
  Chris Quirk, Raymond Mooney and Michel Galley
- "Semantics, Discourse and Statistical Machine Translation"
  Xiong, Deyi, Zhang, Min. 2014.
- "Semantics-based pretranslation for SMT using fuzzy matches"
  Vanallemeersch, Tom, Vandeghinste, Vincent

(note: if no date given, it's ACL 2015)

What's going on in SMT?

continued...

Other related work...

- "Using Syntax-Based Machine Translation to Parse English into Abstract Meaning Representation"
  M. Pust, U. Hermjakob, K. Knight, D. Marcu, J. May, Proc. EMNLP, 2015.
- "An AMR parser for English, French, German, Spanish and Japanese and a new AMR-annotated corpus"
  Lucy Vanderwende, Arul Menezes, and Chris Quirk, Proc. NAACL 2015.
- ...

S2MT Workshop

- 8 submissions, 4 accepted, 4 keynote speeches, panel discussion
  - "Semantic Parsing as, via, and for Machine Translation"
    Percy Liang
  - "Round trips with meaning stopovers"
    Alastair Butler
  - "Learning Multilingual Semantics from Big Data on the Web"
    Gerard de Melo
  - "Integrating Case Frame into Japanese to Chinese Hierarchical Phrase-based Translation Model"
    Jinan Xu, Jiangming Liu, Yufeng Chen, Yujie Zhang, Fang Ming, Shaotong Li
- Are we all talking about the same thing?

Other trends

- NMT
  - Recurrent Neural Network-based Rule Sequence Model for Statistical Machine Translation"
    Heng Yu
  - "Addressing The Rare Word Problem In Neural Machine Translation"
    Thang Luong, Ilya Sutskever, Quoc Le, Oriol Vinyals and Wojciech Zaremba
  - "Encoding Source Language With Convolutional Neural Network For Machine Translation"
    Fandong Meng, Zhengdong Lu, Mingxuan Wang, Hang Li, Wenbin Jiang and Qun Liu
  - ...

Other trends (cntd)

- Evaluation
  - "Improving Evaluation Of Machine Translation Quality Estimation"
    Yvette Graham
  - "Pairwise Neural Machine Translation Evaluation"
    Francisco Guzmán, Shafiq Joty, Lluís Màrquez and Preslav Nakov
  - "Deltableu: A Discriminative Metric For Generation Tasks With Intrinsically Diverse Targets"
    Michel Galley, et. al.
- Reordering
  - "Efficient Top-down Btg Parsing For Machine Translation Preordering"
    Tetsuji Nakagawa

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- "Learning Word Reorderings For Hierarchical Phrase-based Statistical Machine Translation"
  Jingyi Zhang, Masao Utiyama, Eiichro Sumita and Hai Zhao

Where do we fit in?

- We have good semantic representations of (many) sentences
- We know how to do rule-based transfer (and a bit about inducing transfer grammars)
- What about fine-grained word sense distinctions?
- What about sentences without a spanning parse?
- We don't compete well on BLEU (but we do well with human rankings), so how should we evaluate results?

My attempts to fit in...

- I only care about semantics (from and to, not to and from)
- *Bisem* rather than bitext corpora
- We don't have a good bilingual DAG aligner...
- So I try to transform MRS DAGs into alignable components:
  - headed walks
  - subgraph enumeration
  - MrsPaths (no variables, no nodeids, tree structure)
  - aligned subgraphs from bisem (various methods of doing so)
  - ranking alignments (counts, pred alignments, etc.)

Model building

- Map/reduce
- Filtering aligned subgraphs
  - root cvarsort
  - fertility
  - quantifiers
  - stop-graphs

Analyzing subgraphs

"Nearly all dogs bark loudly."

```
- TOP:/H>                                    - TOP:/H>"_bark_v_1_rel"
    "_bark_v_1_rel"(                         - "_bark_v_1_rel":ARG1/NEQ>"_
      :ARG1/NEQ>"_dog_n_1_rel"               - "_dog_n_1_rel"<RSTR/H:_all_
        <RSTR/H:_all_q_rel                   - _all_q_rel:/EQ:"_nearly_x_d
          :/EQ:"_nearly_x_deg_rel" &         - "_bark_v_1_rel"<ARG1/EQ:"_l
      <ARG1/EQ:"_loud_a_1_rel"
    )
```

- What's a "good" subgraph?
  - Any headed subgraph?
  - Only with all or no arguments?
  - Maybe we can learn "cohesive" subgraphs from corpora
- Metrics: depth, distance, diameter, order

Playing with subgraphs

- Merging subgraphs
  - root, frontier, and outpost
  - breadth vs depth merges (i.e. more nodes vs more info)
- Reifying composed subgraphs to MRS (via DMRS)
  - MrsPath can represent more than one DMRS
  - May need to guess cvarsort for nodes

Decoding

- I built a proof-of-concept decoder
- Train:
  - "The dog barks."
  - "The large dog sleeps."
- Test:
  - "The large dog barks."
- It works, but needs some MRS "finishing"
  - Reattach reentrancies
  - Add cvarsorts
  - Add some properties (e.g. TENSE)
- But it needs much more aggressive filtering.

More to do...

- Train larger models with better probabilities
- Build a language model to aid in building transfer models and decoding
- Beam search in decoding
- Evaluate
- ???
- defend!

Discussion

- Call for collaboration
- Consolidate SMT-in-DELPH-IN paradigms
- Shared tasks and publishing (we're not too late to catch the semantics-in-SMT train)